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How can Generative AI be designed and integrated into an ERP Systems Project Management

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List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
ERP	Enterprise Resource Planning
PM	Project Management
CAI	Conversational Artificial Intelligence
GenAI	Generative Artificial Intelligence
BC	Business Central

Abstract

Generative Artificial Intelligence (GenAI) has emerged as a transformative technology which can automate complex tasks, enhancing decision making and improving user experience through natural language interaction. Simultaneously, Enterprise Resource Planning (ERP) systems serve as the digital backbone of current organizations which integrates business processes such as finance, supply chain, human resources and project management. This research focuses on integrating Generative AI into Project Management module of Microsoft Dynamics 365 Business Central to measure and evaluate the potential impact on project managers and project related users. This study explores an implementation approach by developing three integration methods: an Azure OpenAI based approach using Microsoft's Copilot Toolkit, a customized AI assistant solution with direct HTTP connectivity and a native Copilot SDK integration with Azure AI Foundry. Each method was developed using AL (Application Language) extensions using BC development environment, with Claude API explored as an alterative to Azure OpenAI for cost efficiency and performance consideration. The results show that the third method, using native Copilot SDK with Azure AI Foundry, successfully achieved functional integration, enabling conversational AI capabilities within ERP's project management interface. On the other hand, method 1 and 2 encountered technical challenges related to XML parsing and HTTP connectivity, and the successful implementation validates the feasibility of embedding GenAI conversational agents directly into ERP workflows. The findings indicate that GenAI integration could enhance project management cy providing intelligent assistance, context insights and natural language creation through vastly trained data from advanced AI models.

Keywords: Generative Artificial Intelligence, Enterprise Resource Planning, Microsoft Dynamics 365 Business Central, Project Management, Copilot, Large Language Models, Conversational AI.

1 Introduction

The term ERP might be known to many but still there are people who are new to these term even in tech industry. Even some know that ERP is software but not sure what they do in terms of functionality. According to German Tech Giant SAP “Enterprise resource planning (ERP) is a software system that helps organizations streamline their core business processes—including finance, HR, manufacturing, supply chain, sales, and procurement—with a unified view of activity and provides a single source of truth”[1]. It is a business management software system that is designed to manage and streamline an organization’s functions, processes and workflows with automation and integration says another Tech Giant IBM[2]. Simply put, ERP is an integrated software solution that unifies an organization's core processes, eliminates data silos, and enables seamless cross-functional collaboration through automation. Gartner was the first to term ERP in 1990s, but as per Figure 1 ERP initially started as Finance and Accounting tool on Main frame computers and then it became to evolve as an ERP. Organizational operations were completely redesigned about 2010 with the introduction of cloud-based ERP technologies. ERP companies accelerated the shift from on-premises implementations to wholly cloud-based solutions during the pandemic, enabling departments to carry on with their regular operations uninterrupted. By merging several business applications into an integrated solution, ERP software can collect, store, manage and interpret data from cross functional activities. Instead of functioning in silos, different departments can access and share information and at the same time work in a coordinated manner[1].

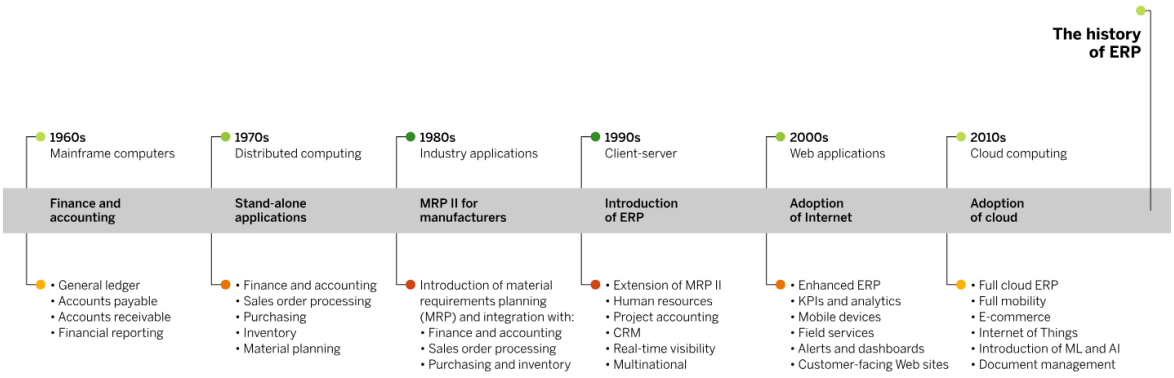


Figure 1: History of ERP, adapted from [3]

The modern ERP systems include interconnected modules to fulfill a company's business process from start to end, and these individual modules can be customized based on the organization's business requirement. There are key modules which contribute to an ERP as a whole, such as financial management, human resources and payroll, supply chain and procurement, manufacturing and production, customer relationship management and also project management. As mentioned earlier, the ERP market is currently focused on Cloud platform, so the global ERP market is expected to grow from USD35 billion in 2023 to USD110 by 2030[4]. Even though cloud-based ERP has grown significantly recently, still the traditional On-premise ERP exists specially in Europe due to data protection laws. This is due to full control over data and the system for an organization. Apart from this traditional deployments there are two other deployment methods. First one is Hybrid ERP, which is a combination of cloud and On-prem method where organizations want to keep the sensitive data on-premise and leveraging the cloud scalability for other functions[5]. The other deployment method is Multi cloud ERP which is not common with Major ERP vendors until now. This method spreads workloads across multiple cloud providers for enhanced redundancy and performance[4]. ERP is expected to grow by 14.4% annually with it future trend of areas like Advanced AI Integration with features like GenAI, augmented intelligence etc[6]. Industry specific solution like pre-configured workflows, compliance features, industry specific analytics etc. and Block chain integration which improves security, transparency and audit trails[7].

Project management is a crucial area which is constantly evolving, driven by new technologies and complex business scenarios[8]. In addition, it is a combination of knowledge, skills, tools, and techniques to complete project activities to achieve specific project goals and objectives[9]. It actually transforms disorganized work into an organized structured and goal-oriented way. Moreover it is set of tools to keep the projects on budget, preventing of overgrowing project scope and finally to achieve the expected results[10]. Project Management was inspired by the Great Pyramid of Giza around 2580-2560 BCE to set an example on how project management should be followed[11]. The are five critical phases in project management, which are initiation, planning, execution, monitoring and controlling, completion/closing[12]. These are often referred to as process groups, which provides a guidance to successfully complete a project from the initial stage. These phases were standardized by Project Management Institute (PMI) and relevant to almost all the projects[13]. Project management best practices highly

contribute to project success as these practices transform disorganized project work into more purposeful journey. When it comes to best practices, there are number of key points which contribute to project success. Clearly defining the project scope & delivery, run a proper initiation, mitigating risk etc[14]. As seen in many industries, project management current and future trend is turning towards the emergence of Artificial Intelligence in Project Management. Fine example would be the dominance of AI through automation, predictive analysis and AI agents. In this paper, AI Agent/Bot will be discussed comprehensively on how it is applied on project management practices[15].

Whether we like it or not, Artificial Intelligence will and currently reshaping the future. From where it started to how it progressed, we have the question of how and how much it's going to change. Since AI is developing so quickly, many businesses are thinking about how they can make use of AI to improve their operations. Nowadays AI can be a tool to assist human work in their day-to-day business functions and in some industries it can totally replace human labor. For example, in healthcare and medical industry AI can enhance diagnostic precision, enabling personalized treatment based on patient records and significantly improve operational efficiency. In 2023 The Food and Drug Administration (FDA) approved 223 AI enabled medical devices when in comparison to just 6 in 2015 which shows the acceleration in AI[16], [17]. On the business transformation stand point, researches show that across 371 companies in different industries they find AI to be improving their organizational performance through knowledge transfer, enhanced innovation and competitive advantage[18]. In an analysis by Frontier in Computer Science shows that AI enablement streamlines business processes, enhance user productivity and optimize decision making. Further AI driven business processes management (BPM) can reduce operational cost and adapt to evolve based on the market conditions by automating processes, enhancing workflow and also to provide intelligent recommendations[19].

2 Literature Review

2.1 Artificial Intelligence

Alan Turing initiated Artificial Intelligence by asking the question “Can machine think?”. Then he further went onto introduce an indirect method called Turing test which led to the beginning of AI[20]. The term Artificial Intelligence formally coined by John McCarthy in 1950s at the Dartmouth Conference in 1956, which was later called as the founding event of Artificial Intelligence[21]. Unlike now, back then it was slow progress in Artificial Intelligence. So after nearly 30 years again the term AI got popular and it further developed into something called Expert Systems. This made everyone to access and use the AI functionality to perform their tasks using machine learning applications[22]. In early 2000s Vanishing Gradient problem was a common hurdle in training deep neural networks (DNN). In 2006 by using Restricted Bolzmann Machines (RBMs) and layer wise pretraining Geoffrey Hinton solved the problem and paved the way for a new era of Artificial Intelligence[23].

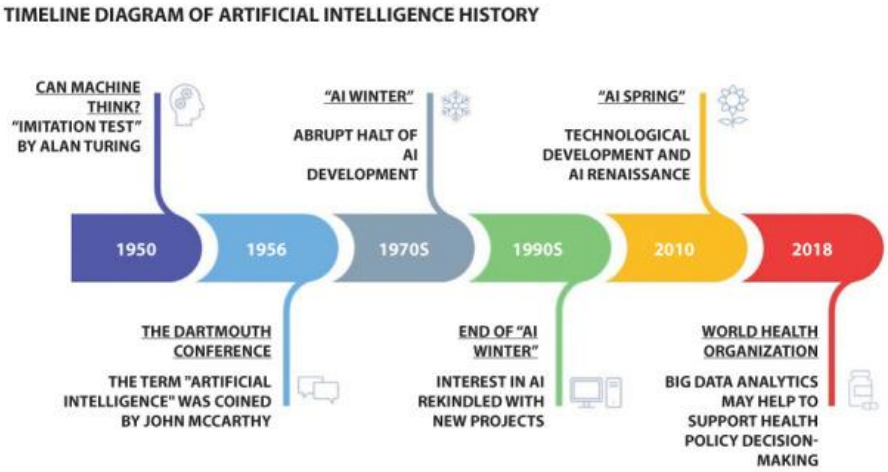


Figure 2: History of Artificial Intelligence. Reproduced from [24]

As per the Figure 2, History of Artificial Intelligence[24] 2010 is where resurgence of AI in medicine happened during this “AI spring”. This is due to the digitalization of all the health data which then led to developments of Big data platforms to create algorithms for AI. This significantly opened up increased funding and technology advancement in AI landscape[24]. Due to this funding organizations like World Health Organizations (WHO) General program of work 2019/2023 and in the European Program of Work (EPW) used Big Data in analysis various medical diagnosis prediction to find clinical outcomes. As per Joost N. Kok, AI has the

ability to mimic intelligent human behavior, it has its challenge in programming computers in a way it can perform numerous tasks to being closer to intelligent[25]. In another study by Guo and Wong, based on Computer Science perspective which focuses on developing hardware and software as a combined systems that can imitate human behavior on decision making to achieve goals[26]. Even though AI technology can imitate complex human skills, it is still far more in achieving it.

2.2 AI Classifications

The classification and categorization of AI is very versatile and it depends on the criteria such as learning method, interaction mode, primary function and Computational capabilities. For example, Avishek Dutta classifies AI by its capabilities or its level of intelligence as Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI) and Artificial Super-Intelligence (ASI)[27]. Further he went on to explain Narrow AI as tasks or domain specific AI which cannot generalize beyond its training and has limitations where it cannot go beyond its trained data or pre-defined rules. This can commonly identify as ChatGPT, Copilot, Alexa etc. General AI which also known as Strong AI can have human level intelligence and has the ability to learn and understand certain tasks, humans can perform. This is still hypothetical or as science fiction where machines can think and reason like humans. Even though this is not practical, it is considered as long term goal of AI. Final one is super-intelligence AI which is described as AI that transcends human intelligence in all disciplines. Again this exists as a science fiction and considered as speculative to till date[27]. Further author of "Taxonomy of Artificial Intelligence" categories in another way as Functional categories which includes Reactive Machines, Limited Memory machines, Theory of Mind systems and Self ware systems[27]. These classifications were initially derived by researchers as the evolution of cognitive sophistication in machines which helped to build the internal complexity of AI agents. Reactive machines are where they do not save any memories and react only to the inputs. They only analyze the current data and do not keep or they do not learn from past data. These can be simple as Chess playing AI which analyzes based on your input on the current state. Limited Memory AI stores temporary data in order to process the current perceptual data to make better decisions. Such examples can be Autonomous vehicle and Chatbots like Siri and Alexa. The other two functional AI categories are pure theoretical based, where Theory of Mind understands emotions, beliefs and intensions of human or other agents and respond. Self

aware AI could understand other emotions and at the same time evaluates and understands its own self like self awareness or self consciousness just like humans or beyond humans[27].

2.3 Major AI Subfields

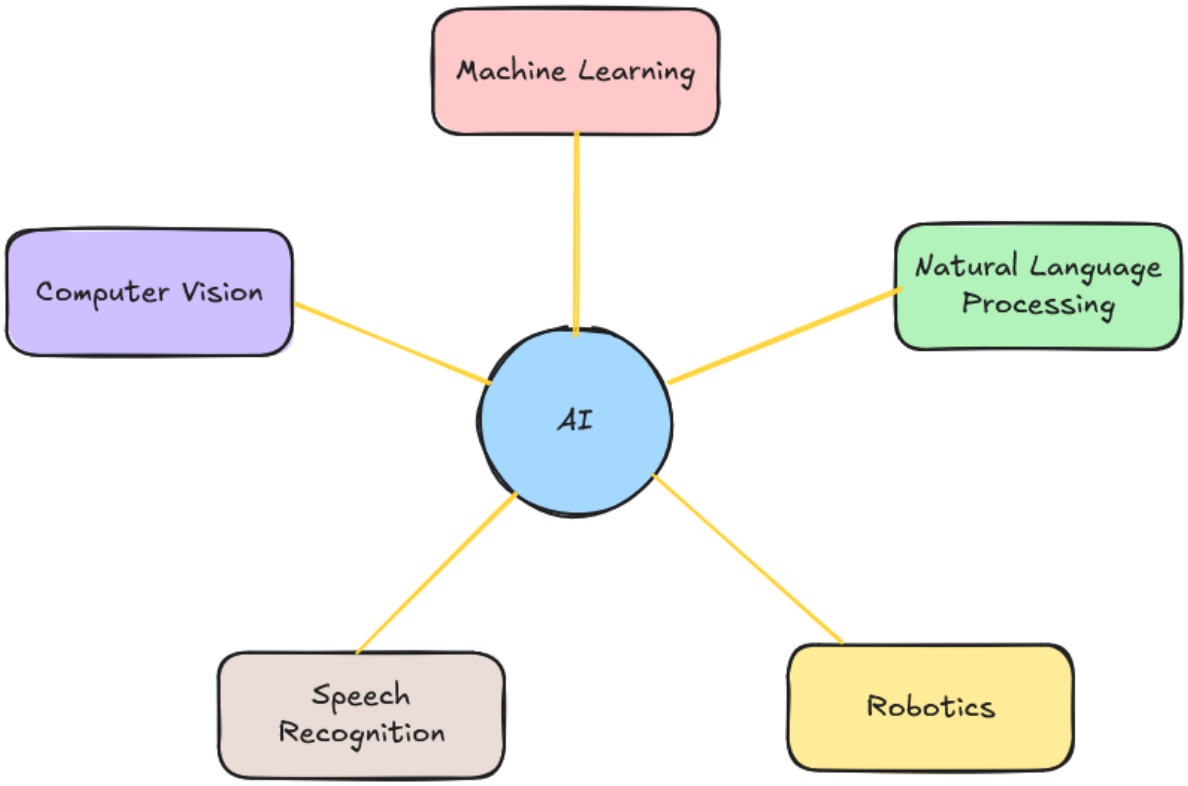


Figure 3: Major AI Sub-types, adapted from [3]

Apart from each categorization, there are 5 major AI sub types. ML focuses on building computer systems that improve through experience automatically. It is also a booming technical field which comprises of computer science and statistics, mostly for artificial intelligence and data science. ML considered as a learning system including computers, humans and organizations through statistical, computational and information theoretic laws studies. ML is generally categorized into supervised learning, unsupervised learning, and reinforced learning. But nowadays there is a blended approach combining these categories are emerging[28]. Further deep learning also comes under ML which uses deep networks as multiple layers of threshold units. For example, in real world ML is used for detecting credit card fraud, which uses labels to classify fraud or not fraud based on the data provided.

Computer Vision empowers computers to see like how humans do. Just like NLP, Machine Learning plays a vital role in process, interpret and derive from images and videos using Convolutional Neural Networks. Computer vision usually classifies image, detect objects, and in some cases it generates images as well. For example, by using deep learning many vision applications helped build self driving cars and for medical imaging[28]. Speech Recognition AI enables machines to convert speech or spoken language into text. This then later used for several purposes and applications. This also called as Automatic Speech Recognition (ASR) or Speech to Text (STT)[29]. ASR comprises several components such as feature extraction to convert speech signal into sequence of acoustic feature vectors, acoustic model contains distinct sound that builds up words in language, language model contains whole set of words in a sequence and finally decoder gets the sound spoken by the human and combines other components to determine text[30]. For example, voice assistants like Siri, Alexa, Gemini converts speech into texts or for further user functions. Robotics is a subtype of an AI which is used in combination of robots, separating from other AI categories due to its physical processing or be able to manipulate objects physically. In other words, a robot should be able to perform tasks that undertake physical actions. Most importantly, robotics integrates all types of AI in reasoning, learning, seeing, hearing, communicating and understanding[3]. In robotics there are categorization or rather forms of robotics. These are not precise categorization, instead based on their functions and nature, it is being grouped. Classical Robots are used as expert systems or rule based AI for standardized tasks. Humanoids are robots that try to mimic human movements like Boston Dynamics Atlas and Big Dog. Drones are well known nowadays for their unmanned photography and military use cases. Autonomous vehicles are also now considered some form of robotics. As per SAE model usually fully automatics operation of the vehicle is considered as robotics. Finally the Autonomous weapon system used in military which is referred to as ‘Killer Robots’. This was introduced in order to limit human damage and losses in conflicts or war. Apart from this there are several use cases like in healthcare assistance, dangerous or difficult tasks, agriculture and agri food etc[3].

Natural Language Processing (NLP) facilitates machines to understand and interpret human language. This is a combination of machine learning, deep learning, statistical modeling with rule based modeling of human language. Usually complexed NLP systems work to recognize, understand, and generate both text. Computational linguistics are the basic foundation for NLP which has two main analytics type called syntactical analysis and semantical analysis.

Syntactical analysis identifies the meaning of the word, phrase, or sentence by using the preprogrammed rules of grammar and Semantical Analysis interpret the meaning of the phrases or words within a sentence using syntactic analysis output[31]. Apart from the components of NLP, there are three approaches which construct the Natural Language Processing. Rule based NLP which is one of the earliest form of NLP and limited in scaling. It uses if then decision tree preprogrammed rules method. Statistical NLP relies on machine learning to extract, label and classify the text or voice data to derive meaning out of it. This introduced a mapping technique of words and grammar to a vector using mathematical or statistical method. Final approach is very well known Deep Learning NLP and this is also the dominant form of approach in NLP. DL models achieve accuracy through huge raw and unstructured data using neural networks model. Most known models are recurrent neural networks (RNN), Google's BERT, GPT, Llama and Claude etc. These models are used in cases like content generation, enhanced search, language translation, automated assistance, data analytics, sentiment analysis, integration with Speech Recognition and many more speech and text generation scenarios[31].

2.4 Generative AI

Advancement in NLP and research surrounding those areas led to empowerment of Generative AI. Further the paper mentions that large language models rely on the advancement in NLP help to achieve communication skills and the ability of image generation models to understand textual requests[31]. In another paper, mentions that Generative models influenced the NLP to become the state of art in several speech and text generation tasks. Specially, DNNs were trained in a way that assign possible next words or phrases in a sentence. Further the author goes onto iterate that key NLP advancement was made due to the Gen AI models introduction which includes question answering, essay writing from prompts, auto sentence completion or Machine translation[32]. These advancement are done though actual data distributions from learned patterns using large training datasets, resulting in new synthetic samples created. Mathematically generative models and discriminative model have very distinct functions. For example, Discriminative represents conditional probability as: $\boldsymbol{P(y|x)}$ and Genertive represents joint probability in: $\boldsymbol{P(x)}$ **or** $\boldsymbol{P(x,y)}$. In short discriminative models focuses on learning the boundary between classes and they answer the primary question of, given the input $\boldsymbol{x}$, what will be the probability it belongs to class $\boldsymbol{y}$. Further these models focus on the classification only and not bothered how the data $\boldsymbol{x}$ was generated. On the other hand, generative models try

to learn the data distribution and they answer the question how the data $\boldsymbol{x}$ looks like overall and try to generate new samples from it[33].

As mentioned in the paper previously, AI typically rooted in 1950s but for generative AI 2015 was the year it started become a significant part of AI. From the emergence of Alpha Go to the landmark release of Chat Generative Pre-trained Transformer which is known as ChatGPT by OpenAI in late 2022 made a significant impact all over the world[34]. ChatGPT became one of the go to topic on Google search due to that fact that Deep belief networks and Restricted Boltzmann Machine were limited in their capacity for generalization[33].Gen AI further advanced due to the introduction of Generative Adversarial Networks (GANs) with the help of two competing neural networks, generator and a discriminator which was explained previously. Later another breakthrough in Gen AI came as the arrival of the Transformer neural network, which utilizes self-attention mechanisms to efficiently process sequential data, bypassing previous architectural constraints of recurrence and convolution[34], [35].

2.4.1 Training Generative AI Models

Initially a massive amount of time and cost is involved when training a generative AI model. Each LLM provider train their models by including large datasets of web pages, newspaper, books, magazines, articles, voice recordings, images and videos. Then humans are assigned to label them or tag them based on the input and decide on the quality of the generated output by the model. Through this process model predicts and learns the expected outputs[36].

A pre-trained model is always considered as an output based on its quality which is approved by the developers. These pre-trained models are often used due to the fact that developing new models will cost lot of money and time. Due to this these pre-trained models are now common for most of the LLM vendors as it has been trained with hundreds of billions of parameters during the training. Moreover, only very few vendors have resources to create new models. These foundational models can transfer knowledge from one domain to another which was learned during the pre-training. In general, a single large model can be fined tuned in order to perform different tasks, rather than training individual models for each tasks or domain from the scratch[37]. Due to this new class of models emerged such as BERT, DALL-E, GPT-3. These models have the capability to reproduce contents like translation, summarization, code generation, image synthesis etc[38]. These models bring key benefits like Broad Transferability which can transfer knowledge from one task to another. Sate of Art Generation which can

produce high quality contents like humans and finally the Efficiency and Cost savings to use the pre-trained models so any user could afford to use it for their day to day tasks[39].

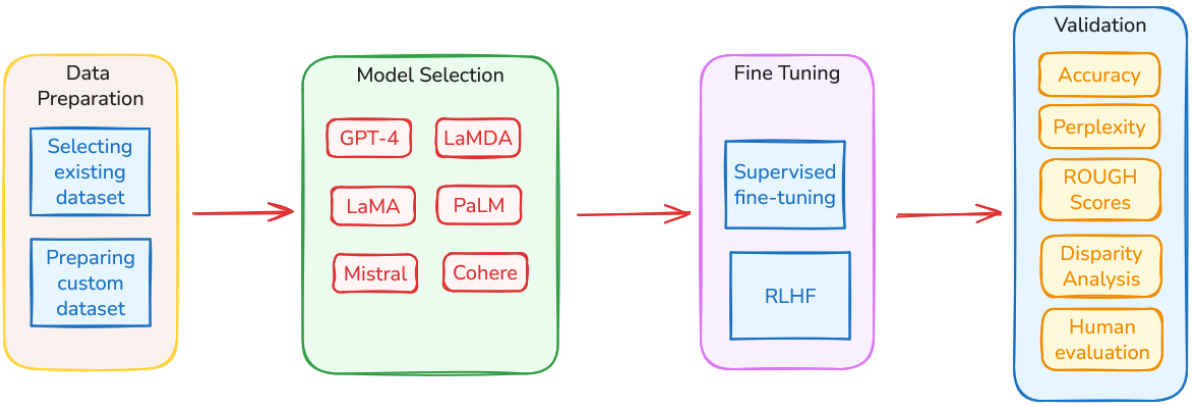


Figure 4: Fine Tuning in Generative AI, adapted from [40]

Fine tuning is the next step in Generative AI models where large pretrained generative model is continuously trained for more specific domain or task until to adapts to the specific use case. Figure 4[40] gives an idea where data preparation and model selection typically the pre-trained data is then fine tuned by supervised fine-tuning and RLHF to get the validated output through different steps[40]. Further author mentions that pretrain then adapt concept can produce desirable outcomes for the task or domain but at the same time the modal size grows significantly when full fine tuning is done. For example, the data can grow upto 175B parameters when updating the weights which results in high performance but also high storage and compute cost is required[40].

2.5 Generative AI's impact on Project Management Practices

2.5.1 Task Automation

The impact of Generative AI in project management responsibilities had led to Automating repetitive tasks in several key areas. As the AI evolves, several studies came up with concepts in keeping up with this change for project management. In a recent study, project managers responsibility were categorized into three key dimensions: automate, assist and augment[41]. These three dimensions provide a framework for automating specific tasks either partial or full AI operated. To illustrate, one of the dimension explain about Automate which will entirely driven by AI and no human intervention will be required as well. These are usually the low coginitive, repetitive and time consuming tasks[41]. Apart from this author of "Automating project lifecycle tasks using Generative AI" clearly mentions that, most of the project lifecycle like project initiation, planning, execution, monitoring and project closure tasks are automated. This significantly reduces the time and effort consumed traditionally by manual project management work[42].

When it comes to Task Automation there are key activities which can be automated. For example, extensive project documentation can be automated and used that time for something which can be used for strategic tasks[41]. Further for reporting, usually project teams need to generate progress reports and updates which can be saved by automating them. In published report it shows that weekly reports could be produced at around 80% faster through NLP based summarization tools[42]. A common activity is the Administrative and Tracking tasks which require project team to enter data and track progress which can be fully automated[43]. Trials has shown that there is 60-75% time saved in task assignment when compared to manual scheduling method[42]. GenAI can automate project planning process by analyzing requirements, constraints and historical project data. Nowadays AI models has the capacity to generate a complete project plans including tasks allocation, dependencies and timelines[43]. For example, GenAI has the ability to convert natural language instructions into a structured, visually appealing plans and Gannt charts[42]. In the domain specific automation, specially in IT project management, GenAI could automate code generation, testing and bud identification[43].

Regardless of Generative AI could automate tasks, the suggestions or ideas did not specifically address ERP systems instead it was more generalized approach for Gen AI in Project

Management. Further it does mentions about Gen AI as a standalone function or integrate with general project management information system platforms like Jira, Asana or Microsoft Project, but it completely unexplored in the area of ERP embedded project management module.

2.5.2 AI Assistant

Emergence of GenAI in project management transformed from mere tool for data analysis to intelligent AI assistant. In order to define its functions as an AI Assistant, it is necessary to have powerful models like ChatGPT which functions on top of neural network architecture like Generative Pre-trained transformer (GPTs), which constantly create new contents and be able to understand sophisticated contexts[41]. To be function as an Assistant, it is necessary to have human oversight or collaboration to define the functions. As per authors one of the dimensions mentioned in the paper, it is partly human and partly AI in streamlining project workflow, scheduling and progress tracking, in order to reduce manual effort but not completely. While augmentation can also be a function of an assistant, project management professional could benefit from completing tasks or activities which previously humans were not able to do it[41]. Further the author discuss about a human AI interaction method which leans towards the copilot mode as a knowledge partner in project management activities[44]. Moreover it provides benefit in human control and supervision and also act as a virtual companion seamlessly integrated into project workflows[45].

Based on what is called pillars of PMI Talent Triangle: ways of working, power skills and business acumen there are several functions which can act as a GenAI assistant. For example, currently AI augmentation can utilize the big data and do analysis or insights which was manually impossible on human effort[41]. In particular, currently GenAI could measure project schedules, guides project timelines and leverages machine learning to forecast allocation on budget and resources[45]. Further the assistant is capable of acting as a cost controller or an analyst to prevent any cost overruns. Additionally it has the capability to predict and forecast on suggesting strategies to mitigating future unknowns[46]. On the contrary, AI assistants are capable of acting as a strategic thinker, problem solver or collaborative leader. For example, by using natural language AI assistants can communicate and collaborate among the team members and also providing suggestions as well[45]. On the leadership and team support perspective GenAI could reduce the workload of project managers and act as a virtual assistant

or team admin. A fine example author mentions is application like Microsoft Copilot and Google Teammate AI which can summarize and suggest follow up actions[44] Most important role the AI assistant could do is be a Virtual Project Manager. Author highlights that this virtual project manager could continuously 24/7 support to global and remote teams regardless of where they are located and what time the project teams work on independently. For example, a paper proposed a PDF driven AI chatbot which was designed to perform PM automation. For instance the chatbot will act as a virtual manager by decoding complex PDFs (interview transcripts and SRS reports) in order to make better decisions[46].

The journals highlighted many areas which AI assistant could help project teams when asked for specific instruction or guidance, but those papers failed to convey the issue of longer conversations where the model loses its older conversations and deviate from the initial context the conversation started. This is particularly a problem for huge enterprise software like ERP, which might require long conversations because there are several modules in ERP which might require old conversations from different modules to give the user expected results.

2.5.3 Bridging Skill and Knowledge Gaps

Integrating Generative AI into project management discipline presents great opportunity in mitigating challenges of skills and knowledge gap. Some research observation suggests that models like ChatGPT-4, bridge the gap for new project management professionals[47]. Studies suggest that GenAI significantly optimizes their performance in both seasoned project management professionals and entry level professionals as well. Most importantly, for the novices it boosts their performance quality and achieve certain tasks which can be comparable to experienced PM professionals without AI assistance[47]. This critically highlights that the expertise gap can be closed between the experienced professionals versus the newer professionals through the assistance of Generative AI. Based on this, GenAI standout as a supplementary tool for this novices to make up for their lack of experience. Most importantly for new project managers could gain a steeper learning curve and faster skill acquisition through leveraging GenAI. Moreover, leveraging GenAI could help the novices to potentially breakdown advanced project management methodologies[47].

Due to the advancement of GenAI, it has the capacity to address complicated project management challenges and vast area of industry knowledge which was outlined by the Project Management Institute (PMI)[48]. The author highlights that well known LLM models such as

ChatGPT 3.5, ChatGPT 4 (Currently Chat GPT 5) and Google Bard (Currently Google Gemini) proven to be proficient in handling questions sourced from PMI certification preparation exams. Further in his study he validates that these models answer the questions and achieve an overall score of 82%. This results confirms that the models can understand the theories, best practices, and methodologies of a complex project management scenarios[49]. Due to the fact that GenAI showcases a string theoretical and some complex practical aspects of the profession, it can perform as advisors to human project managers. This can be invaluable for the project managers because GenAI can support sophisticated decision and also provide reasoning in critical areas of project management. Moreover the author validates that even though the LLM is not finetuned, pre-trained models could do the job as well[49]. Apart from Advisory role and support for PMI certification, Gen AI could achieve near human project deliverables backed by their extensive knowledge in framework and PMBOK guide[48]. For example, the study reveals that GenAI could enhance quality planning and at the time provide insights into risk management which can be complemented by human project managers[50].

Regardless of Generative AI bridging the knowledge gap of project management professionals, still these researches does not address the issue of training of non-technical project management professionals to effectively use generative AI tools. Further research reveals that there is only minimal investigation when it comes to knowledge flow between AI systems and project teams. Even though there are studies which present in closing this knowledge gap but no validated tools or instruments exist to assess AI related skill gaps presented in project management.

2.6 Generative AI's impact on Project Manager's role and Organization

Introduction of Generative AI in project management discipline made a profound shift in traditional process oriented methodologies to value focused frameworks further optimized by the AI. Author quotes as this transformation as "Tsunami of change", where this changes the responsibility of the project manager and the operational method of an organization[41].

2.6.1 Shifting Responsibilities

In one of the research, author introduces the three dimensions in order to support project management professionals in their responsibilities and activities: Automate, Assists and Augment. These frameworks suggests that certain responsibilities can be taken over fully by AI without human intervention and certain responsibilities can be shared with AI. And some

responsibilities like analysis which humans cannot do could be taken over by AI[41]. Just like mentioned before in the paper, automation of GenAI could take over the responsibilities of human project professionals. For example, in the context of documentation and reporting author mentions that extensive documentation, complex reports, preparing project charts and reports could be fully automated[41], [44], [50]. Further minutes of meetings and other summaries required for the project update could be done through NLP easily and this can be save a project assistant or a project coordinators time on huge scale[51]. Next point in the responsibility could be planning and scheduling in project management. Even in this, automation helps to plan project activities, estimate project durations, drafting initial project schedule, utilizing or allocating resources etc[42], [43], [44], [51]. Further a study concretely says that 60-75% of time is saved by employing automated systems compared to manual scheduling[42].

The effect of having GenAI in day to day operational tasks could liberate project managers to more higher level strategic and creative tasks. In general, nowadays this PM role is going towards a phase where the skills are valued which the AI cannot replicate like becoming effective leaders, collaborators and problem solvers[44]. This has opened up areas of strategic domain like acquiring skills in strategic thinking and business acumen. This effects both the project team and the organization where project managers should be able to understand the value chain linking towards business value. In other words, project managers should ensure that project roadmaps must align with organizational goals and its functions[41]. Next point is where the current AI cannot achieve is collaborative leadership and empathy. Humans have the advantage of fostering discussions, resolving conflicts, building trust and cross-functional communications[48]. Studies conclude that current era of AI technology lacks emotional capacity. Intuition and empathy are essential for complex human interaction[51].

2.6.2 Organizational perception

Adoption of generative AI in organizations perspective is about the benefits of efficiency and the deep concern about the security and data integrity. Research confirms that when it comes to recognition of potential in GenAI, project professionals agree on the possible benefits of AI tools particularly in planning 86%, monitoring and controlling 75% and integration 83%. Further they acknowledge that it can provide efficiency, reduce costs and increase in productivity which ultimately free them to concentrate on critical thinking and continuous

improvement. Despite acknowledging the recognition of potential, majority of the professionals around 74% expressed mixed or negative opinions towards Generative AI adoption which mostly referred about the job security and data privacy. Due to this there is a boundary set between tasks AI can automate, and human qualities like trust, empathy and social intelligence. In organization there is always mixed feelings towards AI adoption. For example, higher educated individuals tend to view AI as an opportunity for innovation or opportunity for growth. But whereas those with less education shows greater fear for job loss. More often the younger generation of professionals shows greater enthusiasm towards adopting and using AI in their day to day tasks compared to older workers who are bit skeptical towards job security and data security[50].

2.6.3 Advanced Decision Support System

One great benefit in using GenAI as a decision support system is where it converts complexed projects which has risk management and financial valuation involved to a simple automation task. In other words, GenAI could assist in complex functions like risk prediction, anomaly detection, real time analytics and scenario testing which are mostly absent in traditional PM tools[42]. For example, in risk management GenAi could analyze tasks logs, comment sentiment and industry trends to predict potential risks, suggest mitigation strategies and generate detailed risk assessments focusing on specific threats and opportunities. This presents us with a greater advantage in systematic risk framework which combines AI generated risk plans and human generated risk plans[48]. Advanced models can be used for forecasting and analytics for better decision making. This usually combines historical data and predictive analytics to forecast project timelines, resource requirements and possible bottlenecks. Which creates a proactive measuring way for better decision[43].

While there are plenty of evidence that GenAI automates and augments project management tasks but there is insufficient research about organizational values, cultures and structures changes to AI adoption. Further towards employee perception and training there not much evidence that along with AI adoption upskilling and providing comprehensive training to address the emerging skill gaps. Most of the studies are focused on general project management methodologies but not enough research on specific cases like Agile, Scrum and Kanban to apply GenAI on isolated scenarios.

2.7 Generative AI's impact on Business Processes within ERP

Integrating Generative Artificial Intelligence into Enterprise Resource Planning (ERP) systems marks a significant shift in enterprise technology which changes the traditional ERP rule based automation into an intelligent automation which contains self-learning techniques[52]. Usually ERP systems serves as a digitalized backbone for processes like finance, procurement, sales, project management which is constructed from tens of thousands of processes and billions of code which will be ideal for Gen AI integration towards efficiency and optimization[53]. The transformative impact on GenAI in ERP can be viewed in three critical domains: process automation, specialized data intensive tasks and enhanced user experience.

2.7.1 Business Process Automation

When it comes to automation in business process it usually about business to business transactions, different types of workflows and augmenting general processes within business. These typically includes addressing process limitations related to dynamics and unstructured data within the business to business transaction like emails, contracts, invoices etc. By leveraging GenAI ability like NLP, it can interpret these documents, extract information, and integrating seamlessly into ERP workflows. Through this organization can benefit by increasing operational efficiency, minimizing manual errors and help throughout the decision making processes. Furthermore, GenAI enables the automation of routine, non value adding activities like order entry and invoice matching. Additionally, the continuous learning and adaptability make sure that workflows are constantly optimized[54].

In a study of GenAI in procurement, AI could automate, eliminate or change the substantial existing work of 50 – 80% to self-service models. Further author discusses briefly about GenAI key applications in procurement automation. For example, in ordering and management process GenAI drastically reduces the time dedicated to tasks such as order monitoring by nearly 88% and eliminated processes that take upto 56 minutes of work time. Further tracking of late and unconfirmed order which consumed manual tasks was suggested as well suited for GenAI enhancements. In another business scenario, GenAI could streamline Procure to Pay or Invoice to Pay process which can be totally automated without any human intervention[55].

2.7.2 Complex Data Intensive Tasks

In ERP implementation projects usually involve data intensive tasks frequently, specifically in data conversion and migration. ERP consultants and industry leaders would called this the labor intensive part of ERP implementation which typically consumes time, cost, requires validation checks iteratively and field mappings. With the introduction of GenAI it has revolutionized these intensive tasks into easier methods[56]. Author suggests a framework which enhances efficiency and drastically reducing the need for manual intervention in data conversion. This approach further streamlines the entire data conversion workflow and reduce the time invested in requiring data validation every time the migration happens. For example, in a data extraction and validation process data are first extracted from some legacy system via SQL queries and then exported into formats such as CSV. Later this data is pushed into a machine learning platform to perform initial validation and checking for duplicates and missing values using Pandas libraries. Once the validation is done the model is then used to do data mappings and appropriate data transformations. Finally the converted and transformed data are then inserted into ERP system[56]. By using this framework author concludes that GenAI could make an impact in data conversions and transformations specially when there is a huge data volumes are involved.

2.7.3 ERP User Experience

Regardless of ERP systems expansive functional capabilities there is always usability barriers in either architecture complexity or interface friendliness towards users. To encounter these introducing Conversational AI into the ERP system will make the system more intuitive and make the interface user friendly through natural language. A research paper mentions that Conversational AI systems are capable of understanding, processing and responding to natural language by the help of dialogue management, machine learning and NLP[53]. Due to this non-technical users can typically use ERP by using natural language in their day to day business tasks regardless of how complex the ERP is[52]. In order to achieve this author suggest that integrating conversational agents or virtual assistant directly built into an ERP workflow is the right approach. Further he went onto say that these agents can enhance productivity and reduce training time by providing interactive, intelligent assistance, guiding users through their tasks will minimize and resolve system errors. Moreover these integration could benefit users in navigation and research in an ERP system regardless of the experience the user is having in the

said ERP. For example, Conversational AI(CAI) could augment navigation by avoiding extensive menus in large enterprise software and also the search experience backed by CAI could enhance locating records, files and transactions easily. Furthermore, the author talks about new terminology called Retrieval Augmented Generation (RAG) which CAI overcomes the challenges of retrieving specific information from the huge structured and unstructured ERP data for information retrieval and knowledge management. This RAG focuses on retrieving relevant document passages based on the user's query. This enables real time support, answering queries and troubleshooting issues and errors. Important part the study is where CAI can augment transaction and analytical activities. For instance, CAI can guide employees through complex transactional tasks like placing an order, assisting in core CRUD (Create, Read, Update, Delete) operations. On the other hand or analytical demands, CAI can streamline operational or strategic reports by allowing users to query through natural language[53].

Based on above papers, some research aspects talks about the benefits and usefulness to the user when GenAI is used inside of ERP system but non of the paper discusses about systematic integration of Generative AI into existing ERP Infrastructure. Specially there is not much evidence on legacy systems integration towards the latest AI technology and systems due to the outdated technology lying behind the legacy system make it harder to integrate.

2.8 Integrating Generative AI into ERP

Research papers mainly focus on creation of Generative AI on standalone applications but it is very limited on integrating GenAI into ERP. For example, studies discusses about standalone applications like Asana, Jira, Monday.com or general ERP implementations without GenAI[52], [53], [55]. Further no papers explore area about GenAI conversational agents which can be embedded into ERP system. In another perspective, there are researches identifying PMI knowledge areas but there are lack of research on how GenAI supports ERP native Project Management process. Project management in ERP systems are bit complexed compared to standalone systems as they are closely tight with the organization's financial or resource constraints of an organization. So when GenAI is built around ERP systems, specially for project management finance should be given due diligence but the research does not present anything related to this problem, thus this leaves an area unexplored. Moreover when it comes to control mechanisms, governance and compliance in ERP project management non these

papers barely address the criticalness of those. For instance, no research was conducted on GenAI assisted ERP authorization, workflows, approval processes or segregation of duties. Further no research on role based access control (RBAC) on the usage of Generative AI in ERP. This leaves the ERP in a vulnerable situation where any users in ERP could exploit or manipulate the GenAI in a harmful way.

2.9 Research Question and Hypothesis

The goal of literature review is to provide a theoretical foundation for the research project and to understand certain concepts of the topic with potential readers. The literature review covers the topic Generative AI in ERP and at the same time the methodology and results will answer the primary research question:

How does integrating Generative AI into ERP based Project Management influence project performance, financial control and security outcomes compared with traditional ERP based project management processes?

While answering the primary research question from this research paper, there are sub research questions identified through Literature Reviews which can be concluded with hypothesis and its practical applicability later in the study.

- RQ1: To what extent can an integrated GenAI conversational agent improve efficiency, automation, and usability of native Project Management processes within an ERP system?

  The integration of GenAI conversational agent directly into an ERP's native Project Management module could significantly boosts the project professionals productivity by automating creation of project tasks based on natural language considering the parameters required to automate it. Through these users could drastically reduce time and additional steps in creating those tasks.
- RQ2: How does integration with an ERP affect the accuracy and practical usability of GenAI driven project forecasts, risk assessments and resource optimization recommendations?

  A Pre-trained GenAI model embedded with ERP which can have access to the organizations financial and resource data, could produce more accurate project forecasts and realistic resource availability and optimization recommendations than GenAI models with standalone PM applications that lacks deep financial integration.

- RQ3: How can an embedded Generative AI conversation bot can be used only by the intended users who has access to the bot and relevant allowed data for individuals? Implementing a role based access control (RBAC) or give user permissions through CRUD based on the organization necessity using the ERP native security permissions. This way users can access GenAI through given permissions and also can avoid data leaks and unauthorized access.

3 Methodology

3.1 Tools and Technology required

ERP System

For the ERP system, Microsoft Dynamics 365 Business Central was chosen for this implementation. As per Fig. 5, version 27.0 was chosen for this implementation, where ERP system environment (which user interacts with the system) and development environment (which developers use to further develop the system) versions should be the same.

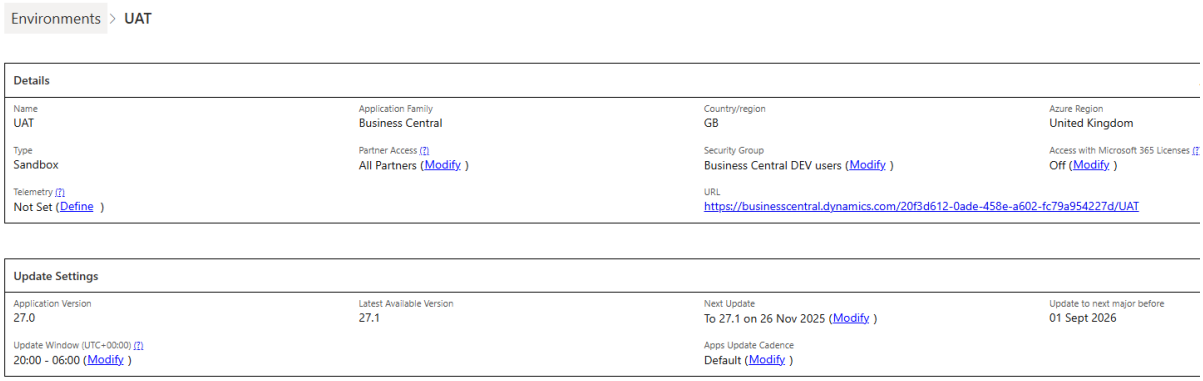


Figure 5: ERP Environment

VS Code and AL Extension

When it comes to development side of ERP it requires VS Code with AL Language extension for Microsoft Dynamics 365 Business Central. Further to match both environments (development and user application) Business Central 2025 release wave 2 was selected and any latest features out of the box will be available for this implementation. Moreover a Git Repo (https://github.com/cliffdani/ALGenAI.git) was created in order to handle any changes and track any development during the implementation.

AI Model

As per standard integration between D365 Business Central and Copilot, Microsoft's own AI model Azure Open AI model was recommended, but for this implementation author wanted to experiment using Claude API (Anthropic). Just like any other LLMs Claude also provides natural language understanding and generation capabilities which enable conversational

interaction with Business Central data. Claude API will utilize its previous generation models like Opus 4, Sonnet 4 and Haiku.

3.2 Architecture Overview

Architecture for this implementation has a combination of different approaches like user interaction, AI processing, business logic and data access. This design covers areas like maintainability, scalability and security. This implementation is based on four main layers. User interface layer provides a chat interface to users to interact with ERP through natural language. Bot application layer handles the coordination of UI and external services, message processing, manage conversation state through Business Central AL code units. AI/LLM Layer interfaces with Claude API for users input through natural language and generating responses based on that. Finally, ERP layer is where users can make use of the LLMs automation capabilities along with standard Business Central functions such as projects tasks, planning lines, resources etc. through native AL development.

3.2.1 Data Flow Architecture



Figure 6: Data Flow

As per the Fig. 6 data flow is initiated from user input through natural language request in the chat interface. For example, “What is the budget status for Project 001”. Then the Bot gathers context by retrieving relevant data from Business Central tables. Along with this AI processing sends the user messages, project context and conversational history to Claude API for further processing. Through response generation, Claude analyses the request and the context and then provides appropriate responses. Finally through response delivery AI response is added to the conversation history and displayed to the user.

3.3 Implementation of Generative Artificial Intelligence

3.3.1 Method 1

The implementation of Generative Artificial Intelligence was implemented in three different methods. First method was followed by an example given by Microsoft Business Central development team on Azure Open AI. The example includes Suggest Jobs/projects in the Business Central Project module which is more relevant to the study. By leveraging one of the early releases of AL Copilot Toolkit, the demonstration shows how to use PromptDialog page type and its special capabilities, as well as how to tweak the parameters in order to call Azure Open AI in AL development environment.

Azure Open AI

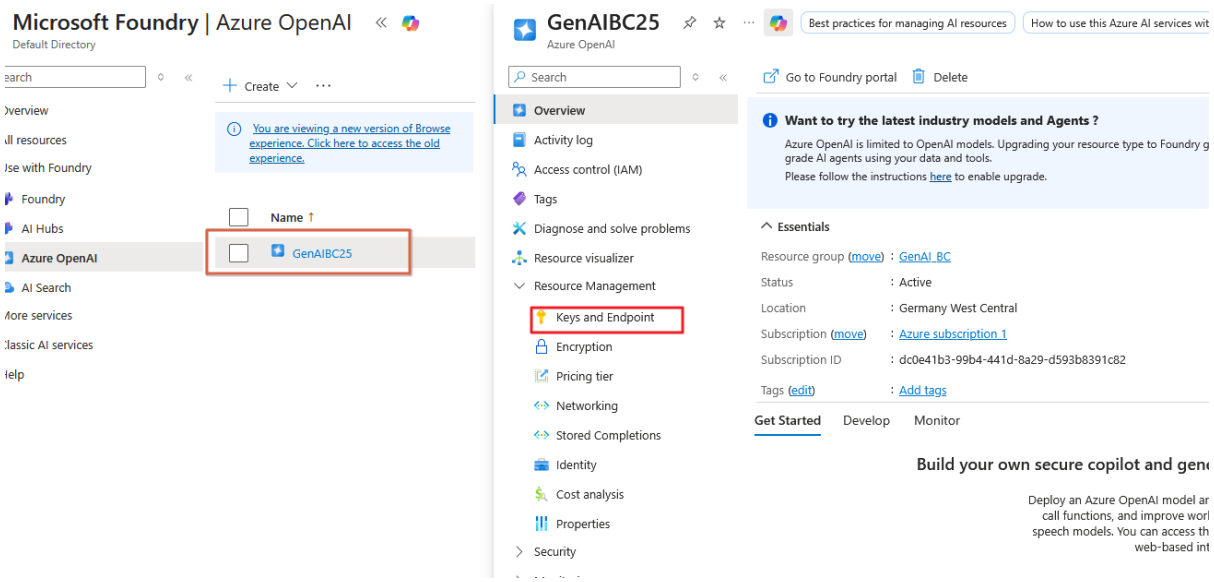


Figure 7: Azure Open AI Setup

As per the Fig. 7 initial setup was to create an Azure Open AI Service. Azure Open AI Service is a platform which hosts several well known models like GPT-5, Dall-e and many other role specific models and make them available through a secure API endpoints. This is useful for any enterprise applications like Business Central to use models directly in its application for various purposes. In this solution Azure OpenAI provides large language models to understand any natural language based questions, reason about the data presented in Business Central and generate human like responses. For this pay as to go subscription account was created and around it resource GenAIBC25 was created. Further as per Fig. 8 “gpt-5-nano” was deployed

as the model for this AI service. From this deployment most important details are the Deployment name, Endpoint URL and API Key for the Business Central AI Service.

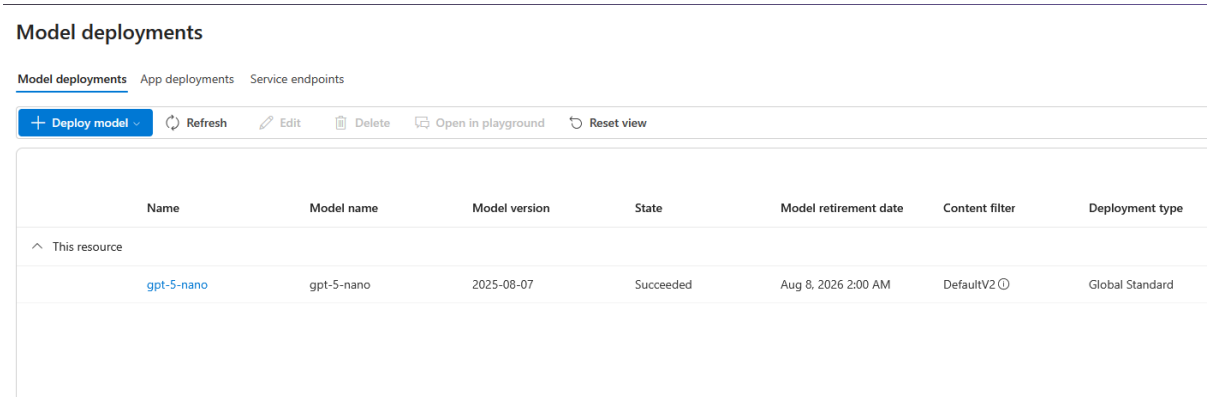


Figure 8: Azure OpenAI Model

Business Central Development

After creating the API endpoint rest of the implementationm is in Business Central environment. Initially there are number of tables, pages and codeunits required to develop and publish into the Business Central. As mentioned before same or compatible user and development environment is important to publish the development. Fig. 9 explain the how the development was done for the Generative AI.

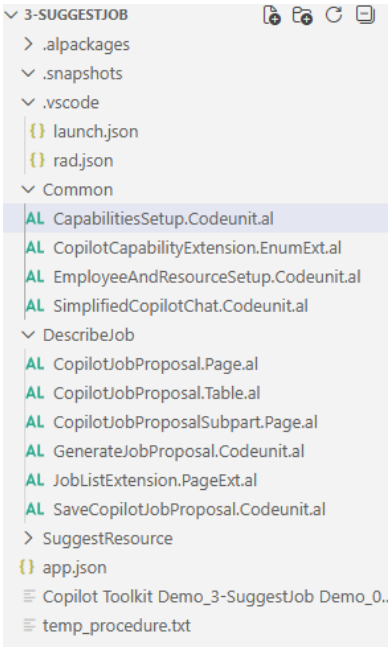


Figure 9: Method 1 BC Development Environment

Configuration Development:

- Capabilities Setup codeunit – this handles setup and registration of Copilot AI capabilities when the extension is installed. This creates a Copilot function called "Describe Project" which can be seen in Fig. 10.

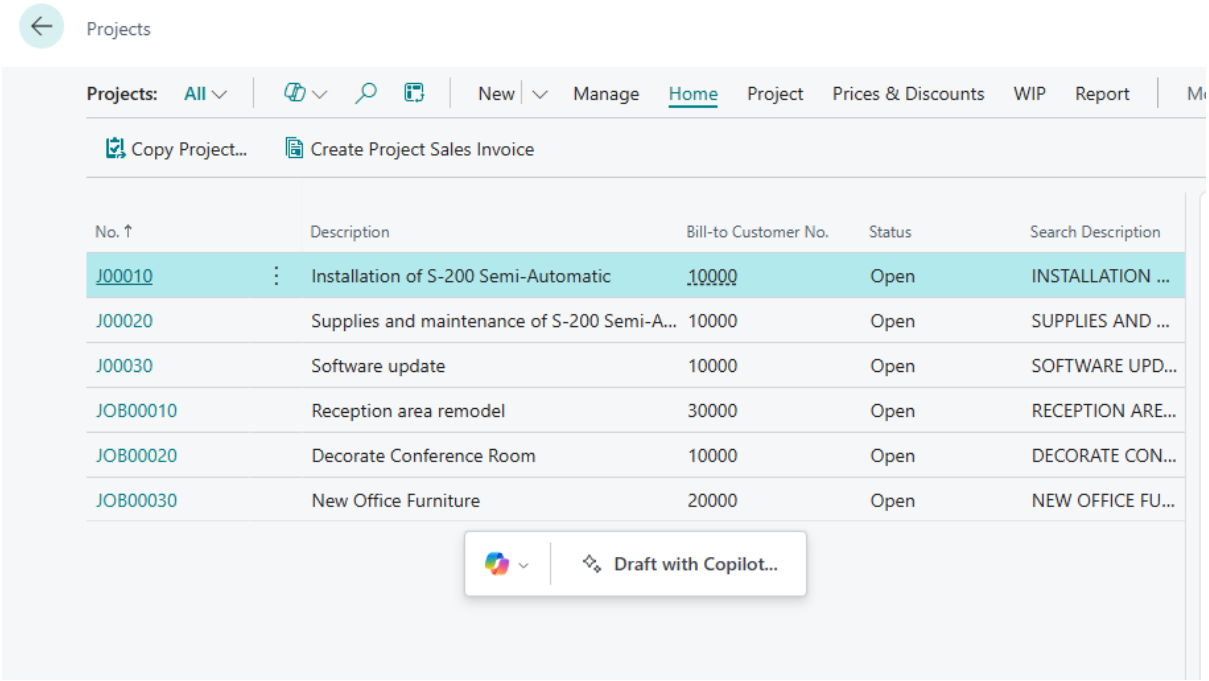


Figure 10: Copilot "Describe Project" function

- Copilot Capability Extension – This basically extends the existing Copilot capability for the developers own development requirement. As shown in Fig. 10 it gives a small intuitive Copilot Icon to use the Conversational Bot.
- Employee and Resource setup codeunit – This creates a sample data to ensure resources are auto created from employees.
- Simplified Copilot Chat codeunit – This is acts as a wrapper around Azure OpenAI API. Its main purpose is to setup authorization with Azure Open AI credentials (Deployment name, Endpoint, API Key) and sends a system prompt plus user prompt to GPT model.

Functional Development:

- Generate Job proposal codeunit – This receives user prompt describing a job/project and then calls SimplifiedCopilotChat.al with a system prompt providing details about project description, customer name and task lists with dates and description. For this Copilot Job Proposal table is used to store AI generated job proposals temporarily (project description, tasks, dates, customer, resource assignments) and Copilot Job Proposal page is used for view those generated data and save those to actual Business Central Projects list.
- Save Copilot Job Proposal codeunit – This persists AI generated proposal into actual project, project tasks and project planning lines records.
- Generate Resource Proposal codeunit and Copilot Resource Proposal codeunit – Similar to project/job generation but focused more on resource allocation like employees, work items, sub contracting etc. Further this suggest which employees are best suited for tasks and help to view those in a page and integrate with project planning lines.

Data Flow summary for Method 1

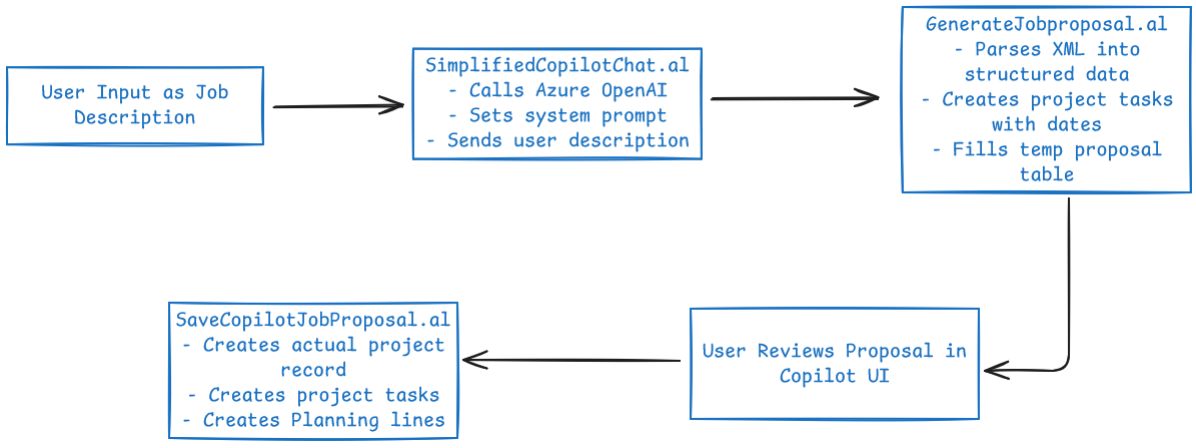


Figure 11: Data Flow for Method 1

3.3.2 Method 2

In this method followed the same procedure for AI model using Azure Open AI but developed the solution simple and masking the API details for further security.

Business Central Development

Just like Method 1, there are key tables, pages and codeunit required for this solution to run. As per Fig. 12 the Generative AI solution was simple with necessary codeunits required to run the Bot. Further rather than using the inbuilt Copilot bot, author created a separate page for the Bot for more flexibility in the solution. The solution acts as a conversation window to interact with the AI where user could see the chat history, ask further questions and send or clear chat based on the situation (Fig. 14). AI Brain/Logic is where the codes connects Business Central to Azure OpenAI to retrieve the specified model in Azure OpenAI. This AI brain gathers project data in Business Central, build the contexts based on the project data, packages it and sent to Azure OpenAI. Finally users will receives AI's response in Business Central Chat terminal from Azure OpenAI model.

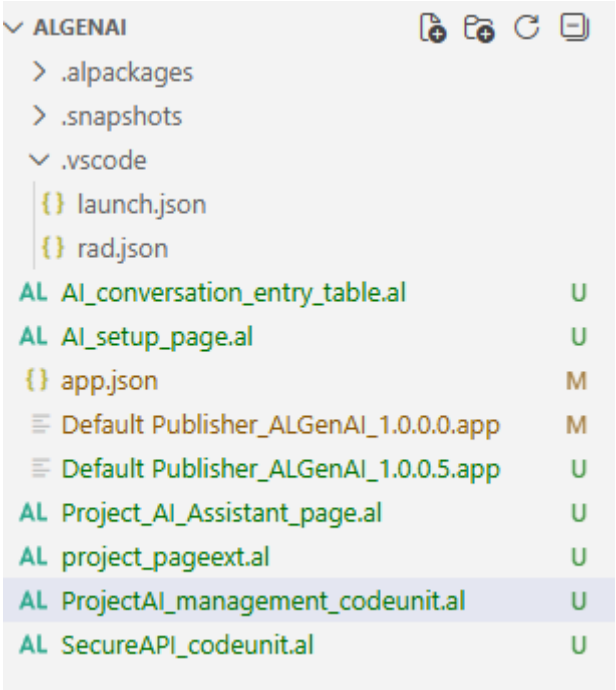


Figure 12: Method 2 BC Development Environment

- AI Conversation Entry table – This is a temporary table to store the conversation messages between user, assistant and system. Here the conversation history is passed to AI requests.
- AI setup page – This is the page where users/admin could configure any API Endpoint, API key and its deployment name. This provides more secure way and also easiest way to maintain and configure API information. Any changes in the endpoint details then developer does not need to hard code it instead changes it in this UI page (Fig.13).

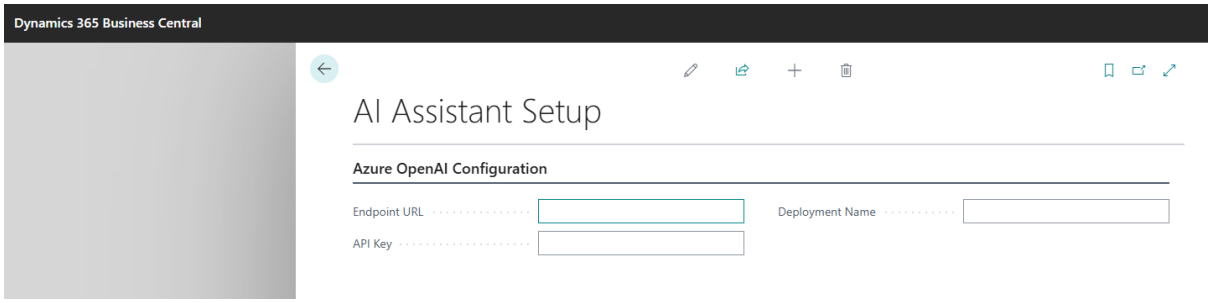


Figure 13: AI Assistant Setup

- Secure API codeunit – A simple wrapper to store and retrieve key in an isolated environment.
- Project page extension – This connects the developed AI Assistant into standard Business Central Project/Job Page.
- Project AI Assistant Page – This provides a chat UI for sending messages to AI and viewing conversations history. This is where the logic of whole conversation bot is happening and what ever the conversation will be held in conversation history table.

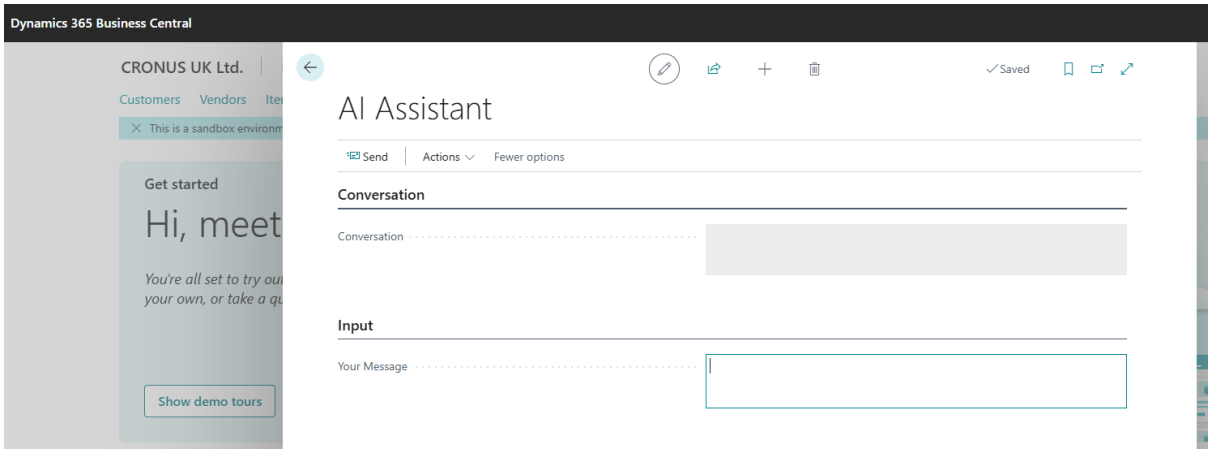


Figure 14: AI Assistant Page

- Project AI management codeunit – This is the main integration layer between Business Central and the AI service, builds request payload, send HTTP calls, parse responses and collects the project context.

Data Flow summary for Method 2

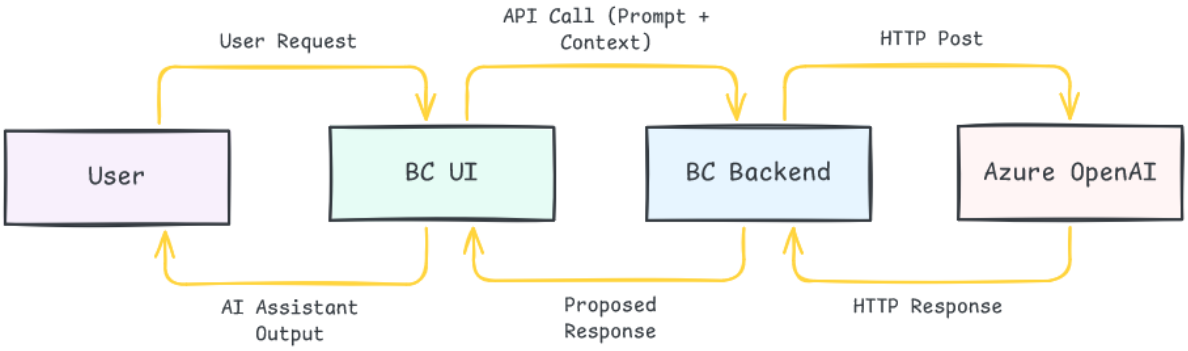


Figure 15: Data Flow for Method 2

3.3.3 Method 3

With method 3 author decided to experiment with Native Copilot software development kit in Business Central and use Azure Foundry for AI model. This method has four architecture layers: configuration layer, user interface layer, integration layer and installation layer. The overall solution is when a user open Project module (Job Card) and clicks the "Ask Copilot" button it opens up a copilot prompt dialog interface that it maintains its awareness of current project context. Then user could leverage their natural language by asking questions which will be based on the opened job tasks, customer information and status. Which will then be packed together with user's question and project context detail in BC and send it to Azure AI Foundry through a secure HTTP client. The AI's response is then parsed and displayed back to the user in real-time, creating a conversational assistance experience to the user.

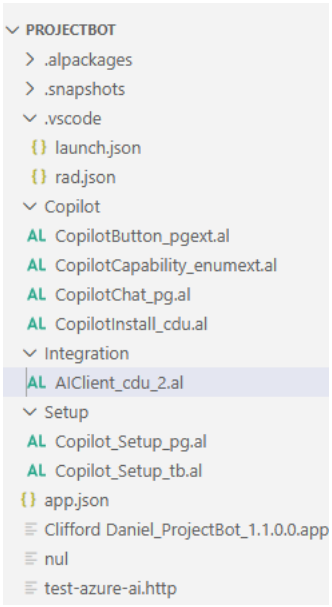


Figure 16: Method 3 BC Development Environment

- Copilot Button page extension – This adds the button for chat interface called "Ask Copilot".

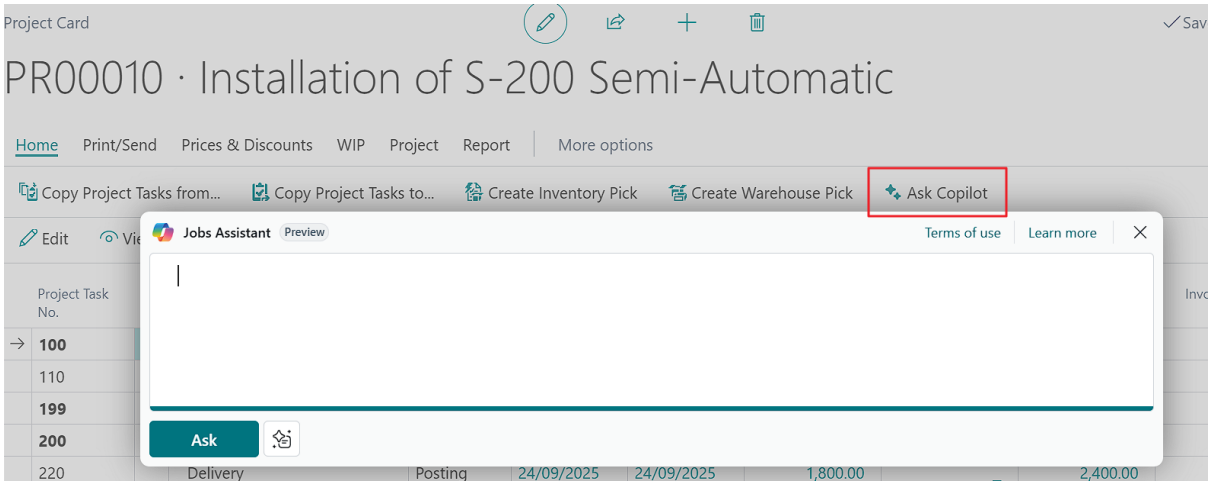


Figure 17: Ask Copilot Button

- Copilot Capability enum extension - As per the Fig. 17 when clicked it "Ask Copilot" it opens a new window for user input.
- Copilot Chat page – This is the UI for prompting AI and showing responses which is associated with the opened Project page. This is mainly to suggest or parse results relevant to the current project context (Fig. 18).

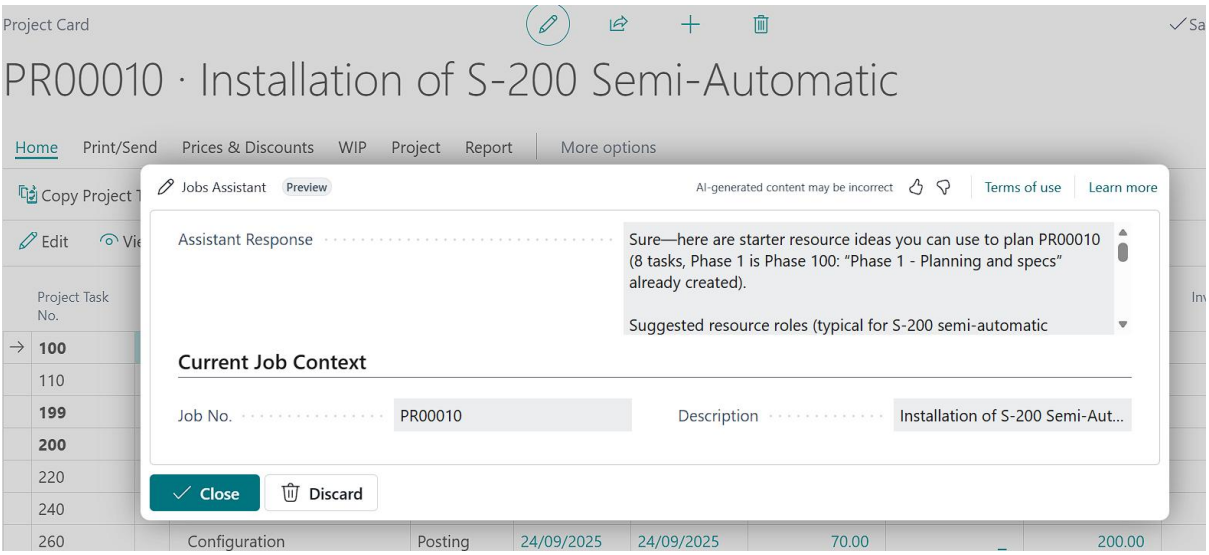


Figure 18: Chat Response from AI model

- Copilot Install codeunit & AI Client codeunit – This two codeunits registers the projectbot with standard copilot inside business central recognizing as a customized solution and sends chat prompts to Azure AI Foundry and parses responses back to BC projectbot UI as per Fig. 8.
- Copilot Setup page and table – This holds the setup page for Azure AI Foundry model's API configuration and masked the API key for security purposes. As per Fig. 19 the configuration details could be changed anytime based on the different models user want to use for their use cases.

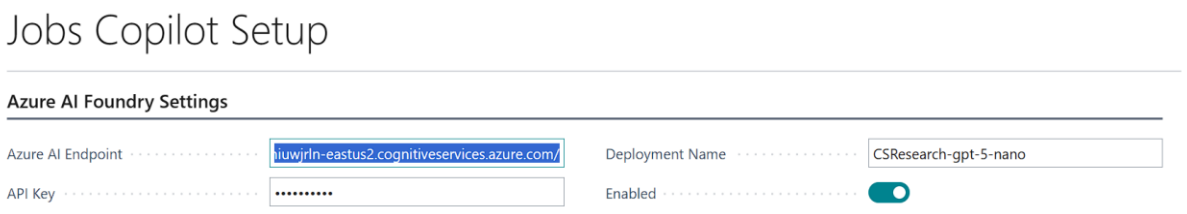


Figure 19: Copilot Setup Page

Data Flow summary for Method 2

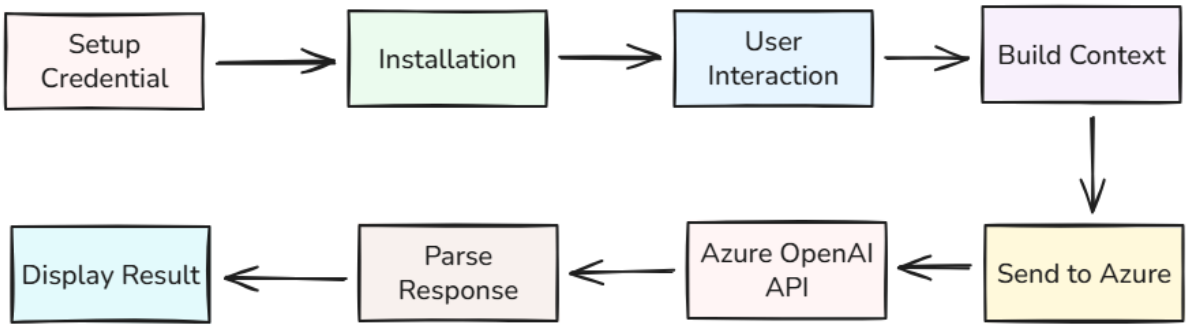


Figure 20: Data Flow for Method 3

4 Results and Conclusion

4.1 Method 1 Results

In method 1 author couldn't achieve the expected results as the Microsoft Dynamics Business Central development team example solution given in the github repository was totally different to the Research Topic of this study and also the example in the Microsoft Github was basic and standard solution. Further the code base was also inserting many data (For example: Creation of resources) before getting actual results from Generative AI. Hence it was focused purely on one particular task for a conversational bot. Nevertheless author wanted to maximize what was given by the Business Central Developers but ended up having error as per Fig. 21. Further inspected regarding the error was due to the XML parsing failure at the root level which indicates the OpenAI model returned non XML content or incomplete XML. This might be due to the model initially used for which is gpt-5 nano, so changed to other previous models which ended up in the same result. Further it could be that Business Central XML buffer cannot parse response because it either starts with reasoning text instead of XML tags, or contains incomplete XML due to token exhaustion.

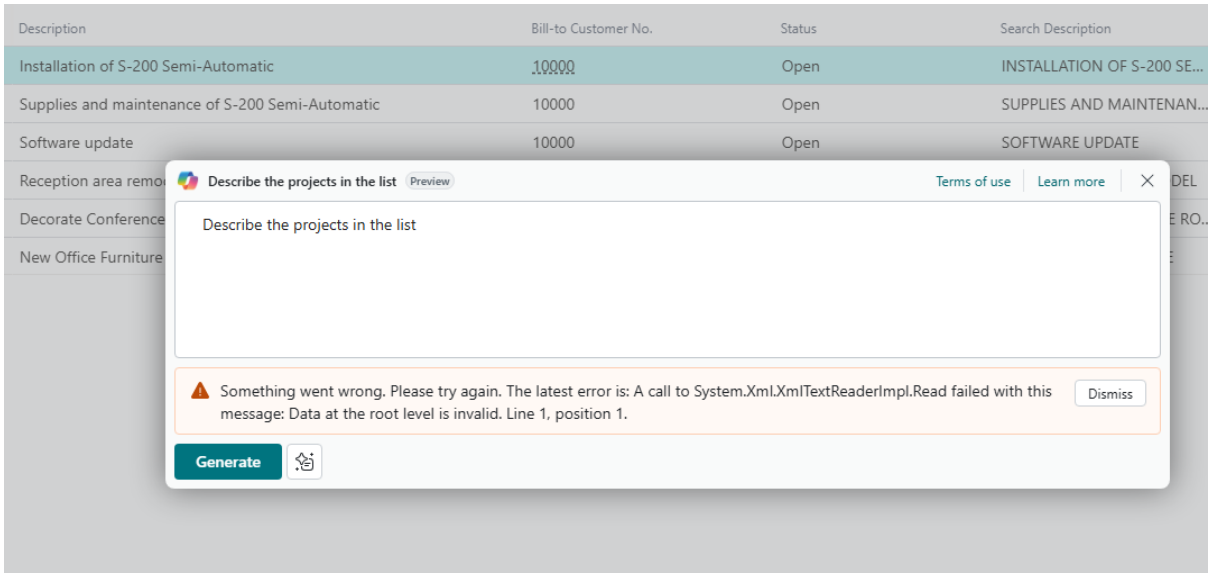


Figure 21: Issue in Method 1

Suggestion:

There might be limitation in this demo sample given by Microsoft's Business Central Development team, but the error could be resolved in following 2 way.

- Fix current Azure OpenAI Implementation through using Azure Foundry instead of Azure OpenAI studio which is an older version compared to Azure Foundry. And deploy simplest models like gpt-4o-mini with setting adequate token generation limit. Further adjusting BC codeunits in providing proper deployment, endpoint and secret key details with adding proper response validation for XML format.
- Another way could be to forego Azure OpenAI and introduced Clause API via direct HTTP calls as Business Central SaaS can have better compatibility with external APIs through proper HTTP client setup.

4.2 Method 2 Results

Since method 1 didn't give expected output, author tried method 2 in a more customized solution. The solution was properly implemented but still there is an issue in connecting to the AI service and the model in Azure Foundry. This error concludes that Business Central extension fails to establish an HTTP connection to Azure Open AI API endpoint which is resulting in the message. This occurs before any response is received from the server, this means that the connection itself is being blocked or failing at the network level instead of being rejected by the API.

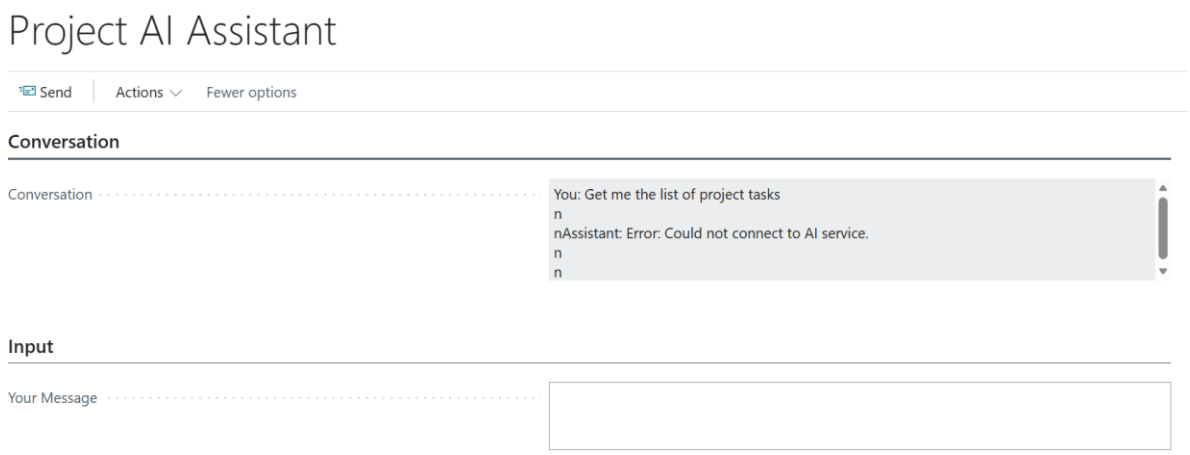


Figure 22: Issue in Method 2

Suggestion:

In order to resolve this issue one way is where trying enabling the HTTP client request inside Business Central manually through the app.json file which permits the extension to make external HTTP calls. And the other way is to tweak the AI setup page by modifying endpoint and API key retrieval procedures to make sure proper credential are being validated in order to connect with Azure AI Foundry model.

4.3 Method 3 Results

This method followed a combination of development and configuration which ended up in successful retrieval of the information from the Azure AI Foundry model. As per Fig. 23, the Job Assistant got a response from AI Model based on the context selected in Business Central page. As can be seen in Got response box, the GenAI model gpt-5-nano provided comprehensive feedback and suggestion for the particular project selected.

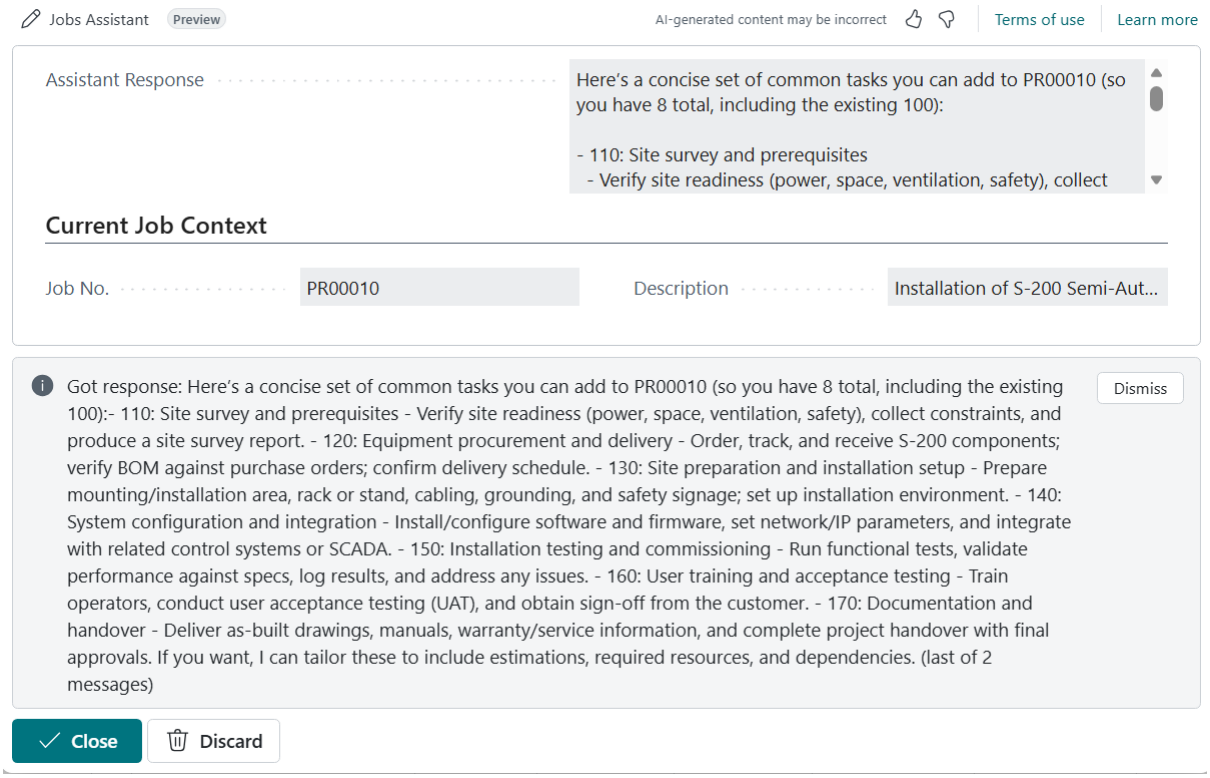


Figure 23: Method 3 Results

4.4 Conclusion and Future recommendation

Even though result for Method 3 was a success and a working solution, it can be further improved to be a human assistant or automate certain tasks in project management portfolio. Furthermore the current solution operates purely as a “Chat Assistant” where it sends contexts and data to Azure AI and displays the text response back to Business Central. It does not perform any database write operations to automate tasks or do allocations of resources. So some recommendations would be to:

- Automating project Task creation – currently the suggest tasks action asks the AI to provide suggestions on common tasks but requires manual creation of those tasks. Instead it can auto generate Project Task Lines based on job type, industry templates, or historical project data and also it could contain template based task libraries that learn from successful past projects which the AI Mode is pretrained with data. Additionally it could further enhanced for intelligent task sequencing with dependency mapping like Task B cannot start until Task A completes.
- Intelligent Resource Allocation – Again the current solution provides advice but does not automatically assign resources based on the project requirements. This could be achieved by skill based resource matching where Business Central has capabilities to add skills and specialization but it does not have any function to make use of that data. With GenAI it could analyze the task requirements and also getting the skills & specialization context from BC and allocate the right resources to the exact projects which it requires certain skills or specialization.

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